VAVR vs Zazr Deep Dive

Season: 7 · Advanced FP · Category: Functional · Level: Advanced

Intent

- Choose the right FP library on Java 25.

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- Classic toolkit vs redesigned sealed toolkit.

When to Use

- Serious FP architecture in Java.
- The design force named in Intent shows up more than once.

When Not to Use

- CRUD apps that don't need effect algebras.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```java
package com.buildwithwaqar.patterns.functional.vavrcompare;

/** Side-by-side Option usage — VAVR (classic) vs Zazr (Java 25 sealed/records). */
public final class VavrVsZazr {
    private VavrVsZazr() {}

    public static String vavrDescribe(io.vavr.control.Option<Integer> opt) {
        return opt.map(n -> "got " + n).getOrElse("missing");
    }

    public static String zazrDescribe(dev.zazr.control.Option<Integer> opt) {
        return switch (opt) {
            case dev.zazr.control.Option.Some(var n) -> "got " + n;
            case dev.zazr.control.Option.None() -> "missing";
        };
    }
}
```

Modern Java 25 / Functional

- Witness types emulate higher-kinded types.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: functional/vavrcompare/
```

Pitfalls

- Encodings are educational — production may use libraries.
- Tagless Final needs Kind encoding without HKTs.

Interview Q&A

- Q: When to pick VAVR vs Zazr Deep Dive? A: When you need testable effect boundaries or pure state threading.
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- Choose the right FP library on Java 25.
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: Classic toolkit vs redesigned sealed toolkit. [0:40] Intent — State the design force in one sentence...